Supplemental Material

This supplement collects technical details that support the main text. We provide derivations and proofs that supports the MTS-CAR model framework assumptions, including the joint prior of the spatial effect, and conjugate full conditionals of the baseline models. We prove the key matrix results (positivity of $D - \rho W$, determinant and Rayleigh characterizations, and identities used to avoid forming large covariances), and verify the algebraic conditions ensuring G is invertible.

Appendix A: Derivation of the joint prior of $\underset{\sim}{\nu} = (\underset{\sim}{\nu}_1,\ \underset{\sim}{\nu}_2)'$

Given the conditional distributions $\underset{\sim}{\nu}_1 \sim \mathcal{N}(\mathbf{0}, \Sigma_1)$, $\underset{\sim}{\nu}_2 \mid \underset{\sim}{\nu}_1 \sim \mathcal{N}(A\underset{\sim}{\nu}_1, \Sigma_2)$, the joint density is $\pi(\underset{\sim}{\nu}_1, \underset{\sim}{\nu}_2) = \pi(\underset{\sim}{\nu}_2 \mid \underset{\sim}{\nu}_1) \cdot \pi(\underset{\sim}{\nu}_1)$. Substituting the normal densities and expanding the quadratic form, we have

$$\pi(\underset{\sim}{\nu}_1, \underset{\sim}{\nu}_2) \propto \exp\left\{-\frac{1}{2}\left[\underset{\sim}{\nu}_2'\Sigma_2^{-1}\underset{\sim}{\nu}_2 - 2\underset{\sim}{\nu}_1'A'\Sigma_2^{-1}\underset{\sim}{\nu}_2 + \underset{\sim}{\nu}_1'A'\Sigma_2^{-1}A\underset{\sim}{\nu}_1 + \underset{\sim}{\nu}_1'\Sigma_1^{-1}\underset{\sim}{\nu}_1\right]\right\}.$$

This can be written as a quadratic form as

$$\propto \exp\left\{-\frac{1}{2}\begin{pmatrix}\underset{\sim}{\nu}_1\\ \underset{\sim}{\nu}_2\end{pmatrix}'\begin{pmatrix}A'\Sigma_2^{-1}A + \Sigma_1^{-1} & -A'\Sigma_2^{-1}\\ -\Sigma_2^{-1}A & \Sigma_2^{-1}\end{pmatrix}\begin{pmatrix}\underset{\sim}{\nu}_1\\ \underset{\sim}{\nu}_2\end{pmatrix}.\right\}$$

Thus, the joint distribution is

$$\underset{\sim}{\nu} = \begin{pmatrix}\underset{\sim}{\nu}_1\\ \underset{\sim}{\nu}_2\end{pmatrix} \sim \mathcal{N}(\mathbf{0}, \mathbf{\Sigma}) \text{ where } \mathbf{\Sigma^{-1}} = \begin{pmatrix}A'\Sigma_2^{-1}A + \Sigma_1^{-1} & -A'\Sigma_2^{-1}\\ -\Sigma_2^{-1}A & \Sigma_2^{-1}\end{pmatrix} \text{ is the precision matrix.}$$

Appendix B. Derivation of $\mathbf{\Sigma}$ from the precision matrix

From Appendix A, define the precision matrix

$$\mathbf{\Sigma^{-1}} = \begin{pmatrix}\Sigma_1^{-1} + A'\Sigma_2^{-1}A & -A'\Sigma_2^{-1}\\ -\Sigma_2^{-1}A & \Sigma_2^{-1}\end{pmatrix} = \begin{pmatrix}B & E\\ F & D\end{pmatrix},$$

where $B = \Sigma_1^{-1} + A'\Sigma_2^{-1}A$, $\quad E = -A'\Sigma_2^{-1}$, $\quad F = -\Sigma_2^{-1}A$, $\quad D = \Sigma_2^{-1}$. We compute $\mathbf{\Sigma}$ using the block-inverse formula with the Schur complement of D. Since D is invertible then

$$S := B - ED^{-1}F, \qquad \mathbf{\Sigma} = \begin{pmatrix}S^{-1} & -S^{-1}ED^{-1}\\ -D^{-1}FS^{-1} & D^{-1} + D^{-1}FS^{-1}ED^{-1}\end{pmatrix}.$$

Compute $ED^{-1}F$. Since $D^{-1} = \Sigma_2$,

$$ED^{-1}F = (-A'\Sigma_2^{-1})\ \Sigma_2\ (-\Sigma_2^{-1}A) = (-A')\ I\ (-\Sigma_2^{-1}A) = A'\Sigma_2^{-1}A.$$

Hence the Schur complement is $S = B - ED^{-1}F = \big(\Sigma_1^{-1} + A'\Sigma_2^{-1}A\big) - A'\Sigma_2^{-1}A = \Sigma_1^{-1}$. Therefore $S^{-1} = \Sigma_1$. Now we compute each block of $\mathbf{\Sigma}$.
Top-left: $S^{-1} = \Sigma_1$.
Top-right: $-S^{-1}ED^{-1} = -\Sigma_1\,(-A'\Sigma_2^{-1})\,\Sigma_2 = -\Sigma_1\,(-A') = \Sigma_1 A'$.
Bottom-left: $-D^{-1}FS^{-1} = -\Sigma_2\,(-\Sigma_2^{-1}A)\,\Sigma_1 = -(-A)\,\Sigma_1 = A\Sigma_1$.
Bottom-right: $D^{-1} + D^{-1}FS^{-1}ED^{-1} = \Sigma_2 + \Sigma_2\,(-\Sigma_2^{-1}A)\,\Sigma_1\,(-A'\Sigma_2^{-1})\,\Sigma_2 = \Sigma_2 + A\Sigma_1 A'$.
Collecting the blocks gives $\mathbf{\Sigma} = \begin{pmatrix} \Sigma_1 & \Sigma_1 A' \\ A\Sigma_1 & \Sigma_2 + A\Sigma_1 A' \end{pmatrix}$, which is the covariance matrix.

Appendix C: Determinant of $\mathbf{\Sigma}$

Observe the factorization

$$\mathbf{\Sigma} = \begin{pmatrix} \Sigma_1 & \Sigma_1 A' \\ A\Sigma_1 & \Sigma_2 + A\Sigma_1 A' \end{pmatrix} = \begin{pmatrix} I & 0 \\ A & I \end{pmatrix} \begin{pmatrix} \Sigma_1 & \Sigma_1 A' \\ 0 & \Sigma_2 \end{pmatrix},$$

Taking determinants,

$$\det(\mathbf{\Sigma}) = \det\begin{pmatrix} I & 0 \\ A & I \end{pmatrix} \det\begin{pmatrix} \Sigma_1 & \Sigma_1 A' \\ 0 & \Sigma_2 \end{pmatrix}.$$

We know that $\det\begin{pmatrix} I & 0 \\ A & I \end{pmatrix} = 1$ and the second factor is block upper-triangular, its determinant is the product of the diagonal element, $\det(\Sigma_1)\det(\Sigma_2)$. Thus

$$\det(\mathbf{\Sigma}) = \det(\Sigma_1)\,\det(\Sigma_2).$$

Since $\Sigma_s = \tau_s^2(D - \rho_s W)^{-1}$,

$$|\mathbf{\Sigma}| = |\tau_s^2(D - \rho_1 W)^{-1}| \cdot |\tau_s^2(D - \rho_2 W)^{-1}| = (\tau_s^2)^\ell |(D - \rho_1 W)^{-1}|\,(\tau_s^2)^\ell |(D - \rho_2 W)^{-1}|.$$

Appendix D: Determinant and Inverse of Σ_s

We define the CAR precision and covariance as $\Sigma_s^{-1} = \frac{1}{\tau_s^2}\big(D - \rho_s W\big)$, We take $\rho_s \in (\frac{1}{\lambda_{(1)}}, \frac{1}{\lambda_{(\ell)}})$ so that $D - \rho_s W$ is positive definite. We introduce the symmetric normalized adjacency $C \equiv D^{-1/2}WD^{-1/2}$. Then $D - \rho_s W = D^{1/2}\big(I - \rho_s\, D^{-1/2}WD^{-1/2}\big)D^{1/2} = D^{1/2}(I - \rho_s C)D^{1/2}$. Therefore, using $(ABC)^{-1} = C^{-1}B^{-1}A^{-1}$, we have

$$\Sigma_s = \tau_s^2\big[\,D^{1/2}(I - \rho_s C)D^{1/2}\big]^{-1} = \tau_s^2\, D^{-1/2}\,(I - \rho_s C)^{-1}\, D^{-1/2}. \tag{12}$$

Since C is real symmetric, there exists an orthogonal U and diagonal $\Lambda = \mathrm{diag}(\lambda_1, \ldots, \lambda_\ell)$ with $C = U\Lambda U'$, $\quad U'U = I$, $\quad \lambda_i \in [-1, 1]$ (for normalized adjacency). Hence $(I - \rho_s C)^{-1} = U\,(I - \rho_s\Lambda)^{-1}\,U'$, where $(I - \rho_s\Lambda)^{-1} = \mathrm{diag}\Big(\frac{1}{1-\rho_s\lambda_1}, \ldots, \frac{1}{1-\rho_s\lambda_\ell}\Big)$. Substituting into (12), we have

$$\Sigma_s \;=\; \tau_s^2\, D^{-1/2}\, U\,(I - \rho_s\Lambda)^{-1}\, U' D^{-1/2}. \tag{13}$$

The matrix $D^{\pm 1/2}$ exists because $d_i > 0$. We require $I - \rho_s C \succ 0$, i.e. $1 - \rho_s \lambda_i > 0$ for all $i = 1, \ldots, \ell$. To find the **determinant**, we start from (13) and use multiplicativity of determinants, $\det(U) = \pm 1$, $\det(D^{-1/2})^2 = \det(D)^{-1}$, and $\det(AB) = \det(A)\det(B)$.

$$\begin{aligned}
\det(\Sigma_s) &= \det(\tau_s^2)\ \det(D^{-1/2})\ \det\big(U(I - \rho_s\Lambda)^{-1}U'\big)\ \det(D^{-1/2}) \\
&= (\tau_s^2)^\ell\ \det(D)^{-1}\ \det\big((I - \rho_s\Lambda)^{-1}\big) \qquad \text{(similarity invariance)} \\
&= (\tau_s^2)^\ell\ \det(D)^{-1}\ \prod_{i=1}^{\ell} \frac{1}{1 - \rho_s\lambda_i}.
\end{aligned}$$

Since D is diagonal, $\det(D) = \prod_{i=1}^{\ell} d_{ii}$, so $\det(D)^{-1} = \prod_{i=1}^{\ell} d_i^{-1}$, thus

$$det(\Sigma_s) \ = \ (\tau_s^2)^\ell \prod_{i=1}^{\ell} d_{ii}^{-1} \prod_{k=1}^{\ell} \big(1 - \rho_s\lambda_k\big)^{-1}, \qquad s = 1, 2$$

The log-determinant form, useful in computation, is

$$\log\det(\Sigma_s) = \ell \log \tau_s^2 - \log\det(D) - \sum_{k=1}^{\ell} \log\big(1 - \rho_s\lambda_k\big).$$

To find the precision, we can invert (12) directly

$$\Sigma_s^{-1} = \frac{1}{\tau_s^2}\, D^{1/2}\, U\,(I - \rho_s\Lambda)\, U' D^{1/2} = \frac{1}{\tau_s^2}\, D^{1/2}\,(I - \rho_s C)\, D^{1/2}$$

Thus, $\Sigma_s^{-1} = \frac{1}{\tau_s^2}\, D^{1/2}\,(I - \rho_s C)\, D^{1/2}, \qquad s = 1, 2$

Appendix E: Computation of G^{-1} and $G^{-1}\underset{\sim}{h}$

We have that $\Sigma_1 = \tau_s^2(D - \rho_s W)^{-1}$, $W = W'$, $a := \underset{\sim}{\nu}_1'\Sigma_2^{-1}\underset{\sim}{\nu}_1$, $b := \underset{\sim}{\nu}_1'\Sigma_2^{-1}W\underset{\sim}{\nu}_1$, $c := \underset{\sim}{\nu}_1'W\Sigma_2^{-1}W\underset{\sim}{\nu}_1$, $d := \underset{\sim}{\nu}_1'\Sigma_2^{-1}\underset{\sim}{\nu}_2$, $e := \underset{\sim}{\nu}_1'W\Sigma_2^{-1}\underset{\sim}{\nu}_2$, for $\underset{\sim}{\nu}_1, \underset{\sim}{\nu}_2 \in \mathbb{R}^L$, then $G = \begin{pmatrix} a & b \\ b & c \end{pmatrix}$, and $\underset{\sim}{h} = \begin{pmatrix} d \\ e \end{pmatrix}$. Since G is 2×2 symmetric, $\Delta := \det(G) = ac - b^2 > 0$, $G^{-1} = \frac{1}{\Delta}\begin{pmatrix} c & -b \\ -b & a \end{pmatrix}$, $G^{-1}\underset{\sim}{h} = \frac{1}{\Delta}\begin{pmatrix} cd - be \\ -bd + ae \end{pmatrix}$. Here, $\Delta = \big(\underset{\sim}{\nu}_1'\Sigma_2^{-1}\underset{\sim}{\nu}_1\big)\big(\underset{\sim}{\nu}_1'W\Sigma_2^{-1}W\underset{\sim}{\nu}_1\big) - \big(\underset{\sim}{\nu}_1'\Sigma_2^{-1}W\underset{\sim}{\nu}_1\big)^2$,

$$G^{-1} = \frac{1}{\Delta}\begin{pmatrix} \underset{\sim}{\nu}_1'W\Sigma_2^{-1}W\underset{\sim}{\nu}_1 & -\underset{\sim}{\nu}_1'\Sigma_2^{-1}W\underset{\sim}{\nu}_1 \\ -\underset{\sim}{\nu}_1'\Sigma_2^{-1}W\underset{\sim}{\nu}_1 & \underset{\sim}{\nu}_1'\Sigma_2^{-1}\underset{\sim}{\nu}_1 \end{pmatrix},$$

and

$$G^{-1}\underset{\sim}{h} = \frac{1}{\Delta}\begin{pmatrix} \big(\underset{\sim}{\nu}_1'W\Sigma_2^{-1}W\underset{\sim}{\nu}_1\big)\big(\underset{\sim}{\nu}_1'\Sigma_2^{-1}\underset{\sim}{\nu}_2\big) - \big(\underset{\sim}{\nu}_1'\Sigma_2^{-1}W\underset{\sim}{\nu}_1\big)\big(\underset{\sim}{\nu}_1'W\Sigma_2^{-1}\underset{\sim}{\nu}_2\big) \\ \big(\underset{\sim}{\nu}_1'\Sigma_2^{-1}\underset{\sim}{\nu}_1\big)\big(\underset{\sim}{\nu}_1'W\Sigma_2^{-1}\underset{\sim}{\nu}_2\big) - \big(\underset{\sim}{\nu}_1'\Sigma_2^{-1}W\underset{\sim}{\nu}_1\big)\big(\underset{\sim}{\nu}_1'\Sigma_2^{-1}\underset{\sim}{\nu}_2\big) \end{pmatrix}.$$

Remarks. (i) Because $\Sigma_2^{-1} = \frac{1}{\tau_1^2}(D - \rho_s W)^{-1}$, every quadratic form contains a common factor τ_1^2, and Δ contains $(1/\tau_1^2)^2$; these cancel between numerator and denominator, so G^{-1} and $G^{-1}\underset{\sim}{h}$ do not depend on τ_2. See below.
(ii) Since $W = W'$ and $\Sigma_2 = \Sigma_2'$, we also have $\underset{\sim}{\nu}_1' W \Sigma_2^{-1} \underset{\sim}{\nu}_2 = (\underset{\sim}{\nu}_2' \Sigma_2^{-1} W \underset{\sim}{\nu}_1)'$, which can be used to check symmetry of the expressions.

Expressing G^{-1} and $G^{-1}\underset{\sim}{h}$ without τ_2^2
We have that $\Sigma_2^{-1} = \frac{1}{\tau_2^2}(D - \rho_2 W) = \frac{1}{\tau_2^2}\widetilde{\Sigma}$, $\widetilde{\Sigma} := (D - \rho_2 W)$. Factor out the common τ_1^2. Since each quadratic form with Σ_2^{-1} equals $(\frac{1}{\tau_2^2})$ times the same form with $\widetilde{\Sigma}$, we can write

$$G = \frac{1}{\tau_2^2} \underbrace{\begin{pmatrix} \underset{\sim}{\nu}_1' \widetilde{\Sigma} \underset{\sim}{\nu}_1 & \underset{\sim}{\nu}_1' \widetilde{\Sigma} W \underset{\sim}{\nu}_1 \\ \underset{\sim}{\nu}_1' \widetilde{\Sigma} W \underset{\sim}{\nu}_1 & \underset{\sim}{\nu}_1' W \widetilde{\Sigma} W \underset{\sim}{\nu}_1 \end{pmatrix}}_{=:\ \widetilde{G}}, \qquad \underset{\sim}{h} = \frac{1}{\tau_2^2} \underbrace{\begin{pmatrix} \underset{\sim}{\nu}_1' \widetilde{\Sigma} \underset{\sim}{\nu}_2 \\ \underset{\sim}{\nu}_1' W \widetilde{\Sigma} \underset{\sim}{\nu}_2 \end{pmatrix}}_{=:\ \widetilde{\underset{\sim}{h}}}.$$

Hence $G^{-1} = \tau_2^2\, \widetilde{G}^{-1}$, $G^{-1}\underset{\sim}{h} = \tau_2^2\, \widetilde{G}^{-1}\Big(\frac{1}{\tau_2^2}\widetilde{\underset{\sim}{h}}\Big) = \widetilde{G}^{-1}\widetilde{\underset{\sim}{h}}$, so $G^{-1}\underset{\sim}{h}$ clearly does not depend on τ_s^2. Moreover, $\det(G) = (\frac{1}{\tau_2^2})^2 \det(\widetilde{G}) \implies G^{-1} = \frac{\operatorname{adj}(G)}{\det(G)} = \frac{\operatorname{adj}(\widetilde{G})}{\det(\widetilde{G})}$, so G^{-1} itself is also independent of τ_2^2.

Let $\widetilde{\Delta} = \big(\underset{\sim}{\nu}_1' \widetilde{\Sigma} \underset{\sim}{\nu}_1\big)\big(\underset{\sim}{\nu}_1' W \widetilde{\Sigma} W \underset{\sim}{\nu}_1\big) - \big(\underset{\sim}{\nu}_1' \widetilde{\Sigma} W \underset{\sim}{\nu}_2\big)^2$. Then

$$G^{-1} = \frac{1}{\widetilde{\Delta}} \begin{pmatrix} \underset{\sim}{\nu}_1' W \widetilde{\Sigma} W \underset{\sim}{\nu}_1 & -\underset{\sim}{\nu}_1' \widetilde{\Sigma} W \underset{\sim}{\nu}_1 \\ -\underset{\sim}{\nu}_1' \widetilde{\Sigma} W \underset{\sim}{\nu}_1 & \underset{\sim}{\nu}_1' \widetilde{\Sigma} \underset{\sim}{\nu}_1 \end{pmatrix}.$$

$$G^{-1}\underset{\sim}{h} = \frac{1}{\widetilde{\Delta}} \begin{pmatrix} \big(\underset{\sim}{\nu}_1' W \widetilde{\Sigma} W \underset{\sim}{\nu}_1\big)\big(\underset{\sim}{\nu}_1' \widetilde{\Sigma} \underset{\sim}{\nu}_2\big) - \big(\underset{\sim}{\nu}_1' \widetilde{\Sigma} W \underset{\sim}{\nu}_1\big)\big(\underset{\sim}{\nu}_1' W \widetilde{\Sigma} \underset{\sim}{\nu}_2\big) \\ \big(\underset{\sim}{\nu}_1' \widetilde{\Sigma} \underset{\sim}{\nu}_1\big)\big(\underset{\sim}{\nu}_1' W \widetilde{\Sigma} \underset{\sim}{\nu}_2\big) - \big(\underset{\sim}{\nu}_1' \widetilde{\Sigma} W \underset{\sim}{\nu}_1\big)\big(\underset{\sim}{\nu}_1' \widetilde{\Sigma} \underset{\sim}{\nu}_2\big) \end{pmatrix}.$$

All occurrences of Σ_2^{-1} have been replaced by $\widetilde{\Sigma} = (D - \rho_2 W)$, so the dependence on τ_2^2 is completely eliminated. Also, Since $D = D'$ (diagonal) and $W = W'$, we have $\tilde{\Sigma} = \tilde{\Sigma}'$. Hence, for any vector $\underset{\sim}{\nu}_1$, $\underset{\sim}{\nu}_1' \tilde{\Sigma} W \underset{\sim}{\nu}_1 = \big(\underset{\sim}{\nu}_1' \tilde{\Sigma} W \underset{\sim}{\nu}_1\big)' = \underset{\sim}{\nu}_1' W' \tilde{\Sigma}' \underset{\sim}{\nu}_1 = \underset{\sim}{\nu}_1' W \tilde{\Sigma} \underset{\sim}{\nu}_1$, so the off–diagonal terms in G^{-1} are equal in magnitude (and opposite in sign) as displayed.

Appendix F: Prove of the algebraic conditions on the scalars

We have $W \in \mathbb{R}^{\ell \times \ell}$ be the unnormalized 0/1 incidence matrix (each row/column sum ≥ 1). Let $D = \operatorname{diag}(d_1, \ldots, d_\ell)$ with $d_i = \sum_j W_{ij}$. For $s \in \{1, 2\}$, $\Sigma_s = \tau_s^2 (D - \rho_s W)^{-1}$, $\frac{1}{\lambda_{(1)}} \leq \rho_s \leq \frac{1}{\lambda_{(\ell)}}$, where $\lambda_{(1)} < \cdots < \lambda_{(\ell)}$ are eigenvalues of $D^{-1}W$. By this choice, $D - \rho_s W$ is positive definite, hence $\Sigma_s \succ 0$, $(s = 1, 2)$. Define $a = \underset{\sim}{\nu}_1'\, \Sigma_2^{-1}\, \underset{\sim}{\nu}_1$, $b = \underset{\sim}{\nu}_1'\, I \Sigma_2^{-1} W'\, \underset{\sim}{\nu}_1$, $c = \underset{\sim}{\nu}_1'\, W' \Sigma_2^{-1} W\, \underset{\sim}{\nu}_1$, and $G = \begin{bmatrix} a & b \\ b & c \end{bmatrix}$. Let $x := \underset{\sim}{\nu}_1$ and $y := W' \underset{\sim}{\nu}_1$ to simplify notation. Because $\Sigma_2^{-1} \succ 0$, the bilinear form $\langle u, v \rangle_{\Sigma_2^{-1}} := u' \Sigma_2^{-1} v$ is an inner product on $\mathbb{R}^\ell$ with induced norm $\|u\|^2_{\Sigma_2^{-1}} = u' \Sigma_2^{-1} u$. In this notation, $a = \|x\|^2_{\Sigma_1}, b = \langle x, y \rangle_{\Sigma_2^{-1}}, c = \|y\|^2_{\Sigma_2^{-1}}$.

(i) Positivity of a. If $x = \underset{\sim}{\nu}_1 \neq 0$, then $a = \|x\|^2_{\Sigma_2^{-1}} > 0$. For our model, we have $\underset{\sim}{\nu}_1 \sim \mathcal{N}(\mathbf{0}, \Sigma_1)$ with $\Sigma_1 \succ 0$, so $\mathbb{P}(\underset{\sim}{\nu}_1 = 0) = 0$; hence $a > 0$ almost surely.

(ii) Nonnegativity/positivity of c. Since $\Sigma_1 \succ 0$, $c = \|y\|^2_{\Sigma_2^{-1}} \geq 0$, with equality $c = 0$ iff $y = W\underset{\sim}{\nu}_1 = 0$. Because every row/column of W has at least one 1, $W \neq 0$, but its nullspace $\ker(W)$ can still be nontrivial in general; thus one cannot assert $c > 0$ for *all* vectors $\underset{\sim}{\nu}_1$. However, under the Gaussian prior $\underset{\sim}{\nu}_1 \sim \mathcal{N}(\mathbf{0}, \Sigma_1)$ (no point mass), $c > 0$ almost surely, since $\mathbb{P}(W\underset{\sim}{\nu}_1 = 0) = 0$.

(iii) Determinant $\det(G) = ac - b^2$. By Cauchy–Schwarz in the inner product $\langle \cdot, \cdot \rangle_{\Sigma_2^{-1}}$,

$$b^2 = \langle x, y \rangle^2_{\Sigma_2^{-1}} \;\leq\; \|x\|^2_{\Sigma_2^{-1}} \, \|y\|^2_{\Sigma_2^{-1}} \;=\; a\,c,$$

hence $\det(G) = ac - b^2 \geq 0$. Equality holds iff x and y are linearly dependent in this metric, which (since $\Sigma_2^{-1} \succ 0$) is equivalent to $W\underset{\sim}{\nu}_1 = t\,\underset{\sim}{\nu}_1$ for some scalar t. Because $\underset{\sim}{\nu}_1$ is continuous and the set $\{v : \; Wv = tv \text{ for some } t\}$ is a finite union of proper subspaces, $\det(G) > 0$ almost surely.

Note *(measure–zero equality cases):* Because $\underset{\sim}{\nu}_1 \sim \mathcal{N}(0, \Sigma_1)$ has a density on $\mathbb{R}^\ell$, the sets $\{\underset{\sim}{\nu}_1 = 0\}$, $\{W'\underset{\sim}{\nu}_1 = 0\}$, and $\{W'\underset{\sim}{\nu}_1 = t\,\underset{\sim}{\nu}_1 \text{ for some } t \in \mathbb{R}\}$ are finite unions of proper linear subspaces and hence have probability 0. On these null sets one may have $a = 0$, $c = 0$, or $b^2 = ac$, respectively. Off these sets, $a > 0$, $c > 0$, and $ac - b^2 > 0$, so $G \succ 0$ almost surely.

Appendix G: Prove that $(D - \rho W)$ is PD for $\rho \in (1/\lambda_{(1)}, 1/\lambda_{(\ell)})$

Given that $W \in \mathbb{R}^{\ell \times \ell}$ is symmetric and $D = \operatorname{diag}(d_1, \ldots, d_\ell)$ with $d_i > 0$ for all i. Define $A(\rho) := D - \rho W$ and let $\lambda_{\min} \leq \cdots \leq \lambda_{\max}$ be the eigenvalues of $D^{-1}W$. To show that

$$A(\rho) \succ 0 \quad \text{for} \quad \rho \in \left(\tfrac{1}{\lambda_{(1)}}, \; \tfrac{1}{\lambda_{(\ell)}}\right),$$

with the usual convention that $1/0 = \pm\infty$ when an endpoint eigenvalue is 0.

Proof

Since $D \succ 0$, its unique symmetric square root $D^{1/2}$ exists and is nonsingular. Set $B := D^{-1/2}\, W\, D^{-1/2}$. Because $W = W^\top$ and $D^{-1/2} = (D^{-1/2})^\top$, B is symmetric. Moreover, $A(\rho) = D - \rho W \;=\; D^{1/2}\big(I - \rho B\big)D^{1/2}$. Congruence by $D^{1/2} \succ 0$ preserves positive definiteness, hence

$$A(\rho) \succ 0 \quad \Longleftrightarrow \quad I - \rho B \succ 0. \tag{14}$$

The matrices B and $D^{-1}W$ are similar $B \;=\; D^{1/2}\,(D^{-1}W)\,D^{-1/2}$, so they share the same multiset of eigenvalues. Denote these real eigenvalues by $\lambda_1, \ldots, \lambda_\ell$ (with any ordering); let $\lambda_{(1)} = \min_i \lambda_i$ and $\lambda_{(\ell)} = \max_i \lambda_i$. Since B is symmetric, write its spectral decomposition $B = Q\Lambda Q^\top$ with Q orthogonal and $\Lambda = \operatorname{diag}(\lambda_1, \ldots, \lambda_\ell)$. Then $I - \rho B \;=\; Q\,(I - \rho\Lambda)\,Q^\top$, so the eigenvalues of $I - \rho B$ are exactly $1 - \rho\lambda_i$ $(i = 1, \ldots, \ell)$. Therefore,

$$I - \rho B \succ 0 \quad \Longleftrightarrow \quad 1 - \rho\lambda_i > 0 \;\text{ for all } i.$$

The system $1-\rho\lambda_i > 0$ for all i is equivalent to the intersection of half-lines

$$\bigcap_{i:\,\lambda_i>0} \{\rho < 1/\lambda_i\} \;\cap\; \bigcap_{i:\,\lambda_i<0} \{\rho > 1/\lambda_i\}.$$

Because $x \mapsto 1/x$ is strictly decreasing on $(0,\infty)$ and on $(-\infty,0)$,

$$\bigcap_{i:\,\lambda_i>0} \{\rho < 1/\lambda_i\} \;=\; \Big\{\rho < \min_{\lambda_i>0} \tfrac{1}{\lambda_i}\Big\} \;=\; \Big\{\rho < \tfrac{1}{\lambda_{\max}}\Big\},$$

and

$$\bigcap_{i:\,\lambda_i<0} \{\rho > 1/\lambda_i\} \;=\; \Big\{\rho > \max_{\lambda_i<0} \tfrac{1}{\lambda_i}\Big\} \;=\; \Big\{\rho > \tfrac{1}{\lambda_{\min}}\Big\}.$$

Combining these,

$$I-\rho B \succ 0 \quad\Longleftrightarrow\quad \rho \in \Big(\tfrac{1}{\lambda_{(1)}},\ \tfrac{1}{\lambda_{(\ell)}}\Big).$$

By (14), $A(\rho) \succ 0$ iff $I-\rho B \succ 0$, proving the claimed equivalence. As such, $A(\rho)$ is invertible.

Note: If some $\lambda_i = 0$, the corresponding inequality $1-\rho\lambda_i > 0$ is vacuous; the interval endpoint becomes infinite ($1/0 = \pm\infty$) and the result above still describes the full set of ρ for which $A(\rho) \succ 0$.

Equivalent determinant and Rayleigh characterizations

1. Using $\det(D-\rho W) = \det(D)\prod_{i=1}^{\ell}(1-\rho\lambda_i)$, positivity of all factors $1-\rho\lambda_i$ (i.e. $\rho \in (1/\lambda_{(1)}, 1/\lambda_{(\ell)})$) is equivalent to $\det(D-\rho W) > 0$ together with positive inertia, hence $A(\rho) \succ 0$.

2. Rayleigh quotients: for any $x \neq 0$,

$$x^\top A(\rho)x = y^\top (I-\rho B)\, y, \qquad y = D^{1/2}x.$$

   Hence

$$\min_{y\neq 0} \frac{y^\top (I-\rho B)\, y}{\|y\|^2} = \min_i \big(1-\rho\,\lambda_i\big),$$

   Therefore $I-\rho B \succ 0$ (and thus $A(\rho) \succ 0$) iff $\min_i(1-\rho\lambda_i) > 0$, i.e., $\rho \in (1/\lambda_{(1)},\ 1/\lambda_{(\ell)})$.

Appedix H: Computation for Single-Series Baseline Model

We outline estimation of the single–series baseline by drawing from the joint posterior in (2), together with the marginal posteriors of the model parameters.

Starting from the joint posterior density, integrate out $\underset{\sim}{\beta}$ to obtain the kernel of a multivariate normal distribution.

$$\int_{\mathbb{R}^2} \exp\left\{-\frac{1}{2\sigma^2}(\underset{\sim}{\beta} - \underset{\sim}{\hat{\beta}}_*)'(X'X+\Sigma_0^{-1})(\underset{\sim}{\beta} - \underset{\sim}{\hat{\beta}}_*)\right\} d\underset{\sim}{\beta} = (2\pi\sigma^2)\cdot |X'X+\Sigma_0^{-1}|^{-1/2}$$

so we have

$$\pi(y_0, \sigma^2 \mid \underset{\sim}{y}) \propto (\frac{1}{\sigma^2})^{(\frac{T+a_0}{2}+1)} |X'X + \Sigma_0^{-1}|^{-\frac{1}{2}} \exp\left\{-\frac{1}{2\sigma^2} b_{(y_0)}\right\}.$$

Also, we integrate out σ^2. We integrate the kernel of an inverse-gamma distribution

$$\int_0^\infty (\frac{1}{\sigma^2})^{(\frac{T+a_0}{2}+1)} \cdot \exp\left\{-\frac{b_{(y_0)}}{\sigma^2}\right\} d\sigma^2 = \Gamma(a) \cdot (b_{(y_0)})^{-(\frac{T+a_0}{2})}.$$

So finally, the marginal posterior of y_0 is

$$\pi(y_0 \mid \underset{\sim}{y}) \propto |X'X + \Sigma_0^{-1}|^{-\frac{1}{2}} \cdot \left[b_{(y_0)}\right]^{-\frac{T+q_0}{2}} \cdot \mathbb{I}_{[c_0,d_0]}(y_0),$$

that is $y_0 \in [c_0, d_0]$.

From the joint posterior distrubution, $\underset{\sim}{\beta}$ appears only in the term in the first exponent of (1), which is a kernel of Gaussian, so we have that:

$$\underset{\sim}{\beta} \mid y_0, \underset{\sim}{y}, \sigma^2 \sim \mathcal{N}(\underset{\sim}{\hat{\beta}}_*, \sigma^2(X'X + \Sigma_0^{-1})^{-1}).$$

After marginalizing $\underset{\sim}{\beta}$ from the joint posterior (as above) and collecting terms in $1/\sigma^2$, the resulting kernel is that of an Inverse–Gamma distribution. Consequently, the conditional posterior of σ^2 is

$$\sigma^2 \mid y_0, \underset{\sim}{y} \sim \text{Inverse-Gamma}\left(\frac{T+a_0}{2}, \frac{b_{(y_0)}}{2}\right).$$

We adopt a hybrid posterior sampling scheme that couples grid–based marginalization with conditional simulation. Specifically, we generate 1,000 posterior draws of $\left(y_0, \underset{\sim}{\beta}, \sigma^2\right)$. This direct random-sampling procedure (non-MCMC) does not require convergence monitoring.

Appedix I: Computation for Multiple series Model

We show how to fit the joint posterior distribution in (11).
We have shown the joint posterior (11) in the main manuscript as

$$\begin{aligned}\pi\left(\underset{\sim}{m}_0, \underset{\sim}{\beta}_i, \sigma^2, \tau_\mu, \tau_b, \rho, y_{0i} \mid \underset{\sim}{y}_i\right) \propto (\sigma^2)^{-(\ell T/2+1)} \exp\left(-\frac{1}{2\sigma^2}\sum_{i=1}^{\ell} \|\underset{\sim}{y}_i - X_i\underset{\sim}{\beta}_i\|^2\right) \\ \times |\hat{\Sigma}_{(\tau_\mu,\tau_b,\rho)}|^{-\ell/2} \exp\left(-\frac{1}{2}\sum_{i=1}^{\ell}(\underset{\sim}{\beta}_i - \underset{\sim}{m}_0)'\hat{\Sigma}^{-1}_{(\tau_\mu,\tau_b,\rho)}(\underset{\sim}{\beta}_i - \underset{\sim}{m}_0)\right) \\ \times \mathbf{1}\{0 < \tau_\mu < 1\}\,\mathbf{1}\{0 < \tau_b < 1\}\,\mathbf{1}\{-1 < \rho < 1\}. \qquad (11)\end{aligned}$$

For simplicity, moving forward, we write $\hat{\Sigma}_{(\tau_\mu,\tau_b,\rho)}$ simply as $\hat{\Sigma}$.

Conditional distributions (CPDs) for Gibbs sampling

Block update for (μ_i, b_i)

Each series has a random intercept μ_i and random AR slope b_i that enter the likelihood jointly. Sampling $\underset{\sim}{\beta}_i = (\mu_i, b_i)$ together reduces posterior dependence between them (induced both by the likelihood and by the correlated bivariate prior), improving mixing relative to univariate updates.

Collecting the terms involving $\underset{\sim}{\beta}_i$ from (11), and after some straightforward algebra, the CPD is obtained as

$$\underset{\sim}{\beta}_i \mid \underset{\sim}{m}_0, \Sigma, \sigma^2, \underset{\sim}{y}_i \sim \mathcal{N}_2(m_i, V_i),$$

with

$$V_i = \left(\hat{\Sigma}^{-1} + \frac{1}{\sigma^2} X_i^\top X_i\right)^{-1}, \qquad m_i = V_i \left(\Sigma^{-1}\theta + \frac{1}{\sigma^2} X_i^\top \mathbf{y}_i\right).$$

This block controls how each series-specific trajectory is explained by its own intercept and persistence. Because $\rho \neq 0$ and the data term couples μ_i and b_i through $X_i' X_i$, blocked sampling substantially reduces random-walk behavior in the Markov chain.

Block update for (θ_1, θ_2)

The hypermeans θ_1, θ_2 pool information across series. They are updated jointly because the bivariate prior links μ_i and b_i and because $\underset{\sim}{m}_0$ is two-dimensional.

Given $\underset{\sim}{\beta}_i$ and $\hat{\Sigma}$, the prior model is $\beta_i \sim \mathcal{N}_2(\underset{\sim}{m}_0, \hat{\Sigma})$. With a flat prior, $\pi(\theta_1, \theta_2) \propto 1$, we define the mean, $\underset{\sim}{\bar{\beta}}$, and after some algebra, the conditional posterior is

$$\underset{\sim}{m}_0 \mid \underset{\sim}{\beta}_i, \hat{\Sigma} \sim \mathcal{N}_2\left(\underset{\sim}{\bar{\beta}},\ \frac{1}{\ell}\hat{\Sigma}\right), \qquad \underset{\sim}{\bar{\beta}} := \frac{1}{I}\sum_{i=1}^{\ell} \underset{\sim}{\beta}_i.$$

This update determines the overall population location around which (μ_i, b_i) shrink. When ℓ is large, the posterior variance $(1/\ell)\hat{\Sigma}$ becomes small and $\underset{\sim}{m}_0$ is tightly identified, increasing the strength of pooling across series.

Update for σ^2

The observation variance σ^2 calibrates the within-series noise level. It has a conjugate update, making it efficient to sample exactly.

Collecting the terms involving σ^2 in (11), and defining the residual sum of squares

$$\text{SSE} := \sum_{i=1}^{\ell}\sum_{t=1}^{T} (y_{it} - \mu_i - b_i y_{i,t-1})^2 = \sum_{i=1}^{\ell} \|\underset{\sim}{y}_i - X_i \underset{\sim}{\beta}_i\|^2.$$

after some algebra, the CPD is

$$\sigma^2 \mid \underset{\sim}{\beta}_i, \underset{\sim}{m}_0, y_{i0}, \underset{\sim}{y}_i \sim \text{Inv-Gamma}\left(\frac{\ell T}{2},\ \frac{\text{SSE}}{2}\right).$$

Larger σ^2 weakens the data likelihood and allows greater dispersion in the latent trajectories, whereas smaller σ^2 forces (μ_i, b_i) to fit the observed time series more closely.

Block grid update for (τ_μ, τ_b)

The transformed variance parameters $(\tau_\mu, \tau_b) \in (0,1)^2$ yield compact support and uniform priors, enabling stable grid sampling. Because $\hat{\Sigma}$ depends on both through σ_μ^2, σ_b^2 and is coupled via ρ, updating (τ_μ, τ_b) jointly can improve mixing.

Let $\delta_i := \underset{\sim}{\beta}_i - \underset{\sim}{m}_0$, $i = 1, \ldots, \ell$, and consider the conditional posterior of (τ_μ, τ_b) from (11),

$$\pi(\tau_\mu, \tau_b \mid \underset{\sim}{\beta}_i, \underset{\sim}{\theta}, \rho) \propto |\hat{\Sigma}|^{-\ell/2} \exp\left(-\frac{1}{2} \sum_{i=1}^{\ell} \delta_i' \hat{\Sigma}^{-1} \delta_i \right), \quad 0 < \tau_\mu < 1,\ 0 < \tau_b < 1.$$

Using the transformation defined earlier, we have

$$\sigma_\mu^2 = \frac{1-\tau_\mu}{\tau_\mu}, \qquad \sigma_b^2 = \frac{1-\tau_b}{\tau_b}.$$

The covariance matrix $\hat{\Sigma}$ is now

$$\hat{\Sigma} = \begin{pmatrix} \sigma_\mu^2 & \rho\sqrt{\sigma_\mu^2 \sigma_b^2} \\ \rho\sqrt{\sigma_\mu^2 \sigma_b^2} & \sigma_b^2 \end{pmatrix} = \begin{pmatrix} \frac{1-\tau_\mu}{\tau_\mu} & \rho\sqrt{\frac{1-\tau_\mu}{\tau_\mu}\frac{1-\tau_b}{\tau_b}} \\ \rho\sqrt{\frac{1-\tau_\mu}{\tau_\mu}\frac{1-\tau_b}{\tau_b}} & \frac{1-\tau_b}{\tau_b} \end{pmatrix}.$$

Its determinant and inverse are

$$|\hat{\Sigma}| = \frac{(1-\tau_\mu)(1-\tau_b)}{\tau_\mu \tau_b}(1-\rho^2),$$

$$\hat{\Sigma}^{-1} = \frac{1}{1-\rho^2} \begin{pmatrix} \dfrac{\tau_\mu}{1-\tau_\mu} & -\rho\sqrt{\dfrac{\tau_\mu \tau_b}{(1-\tau_\mu)(1-\tau_b)}} \\ -\rho\sqrt{\dfrac{\tau_\mu \tau_b}{(1-\tau_\mu)(1-\tau_b)}} & \dfrac{\tau_b}{1-\tau_b} \end{pmatrix}.$$

Writing $\delta_i = (\delta_{\mu i}, \delta_{bi})'$, the quadratic form becomes

$$\delta_i' \hat{\Sigma}^{-1} \delta_i = \frac{1}{1-\rho^2} \left(\frac{\tau_\mu}{1-\tau_\mu} \delta_{\mu i}^2 - 2\rho\sqrt{\frac{\tau_\mu \tau_b}{(1-\tau_\mu)(1-\tau_b)}}\, \delta_{\mu i}\delta_{bi} + \frac{\tau_b}{1-\tau_b}\delta_{bi}^2 \right).$$

Substituting this expression and the determinant into the conditional posterior of (τ_μ, τ_b) yields an explicit log-posterior kernel for (τ_μ, τ_b), which is evaluated on a two-dimensional grid and updated jointly in a grid-based Gibbs step.

The parameters (τ_μ, τ_b) govern cross-series heterogeneity. Small values of τ correspond to large variances $\sigma^2 = (1-\tau)/\tau$, allowing substantial series-specific variation in (μ_i, b_i), while large τ enforces strong pooling toward the population mean $\underset{\sim}{m}_0$. Joint updating of (τ_μ, τ_b) is particularly important when $|\rho|$ is large, as the posterior dependence between these parameters is then strong.

Grid update for ρ

The correlation ρ governs prior dependence between μ_i and b_i across series, capturing whether higher intercepts tend to coincide with higher (or lower) AR persistence. Because $\rho \in (-1, 1)$ and enters nonlinearly through $|\hat{\Sigma}|$ and $\hat{\Sigma}^{-1}$ obtained earlier, a grid update is convenient and robust.

From (11), after a straightforward algebra, the full conditional is

$$\pi(\rho \mid \tau_\mu, \tau_b, \underset{\sim}{\beta}_i, \underset{\sim}{m}_0) \propto (1-\rho^2)^{-\ell/2} \exp\left(-\frac{1}{2}\sum_{i=1}^{\ell} \delta_i' \hat{\Sigma}^{-1} \delta_i\right) \mathbf{1}\{-1 < \rho < 1\}.$$

Positive ρ encourages μ_i and b_i to co-vary positively across series, while negative ρ implies a tradeoff (e.g., higher intercepts associate with lower persistence). Estimating ρ can substantially improve fit and pooling structure when intercept and persistence are empirically linked.

Update for y_{i0}

Since y_{i0} is unknown (latent), it must be updated because it appears in the likelihood through the first transition equation for y_{i1} via the lagged predictor. Sampling y_{i0} propagates uncertainty about the initial condition into inference on (μ_i, b_i).

For each series i, the only factors involving y_{i0} are the prior

$$y_{i0} \sim \mathcal{N}(\mu_0, \sigma_0^2)$$

and the $t = 1$ likelihood contribution

$$y_{i1} \mid y_{i0}, \mu_i, b_i, \sigma^2 \sim \mathcal{N}(\mu_i + b_i y_{i0},\, \sigma^2).$$

Thus, up to proportionality the conditional posterior from (12) is

$$p(y_{i0} \mid y_{i1}, \mu_i, b_i, \sigma^2, \mu_0, \sigma_0^2) \propto \exp\left\{-\frac{(y_{i0}-\mu_0)^2}{2\sigma_0^2}\right\} \cdot \exp\left\{-\frac{(y_{i1}-\mu_i-b_i y_{i0})^2}{2\sigma^2}\right\}.$$

This is Gaussian in y_{i0}. After some algebra on completing the square, the CPD yields

$$y_{i0} \mid y_{i1}, \mu_i, b_i, \sigma^2, \mu_0, \sigma_0^2 \sim \mathcal{N}(m_{0i}, V_{0i}),$$

with precision and mean

$$V_{0i} = \left(\frac{1}{\sigma_0^2} + \frac{b_i^2}{\sigma^2}\right)^{-1}, \qquad m_{0i} = V_{0i}\left(\frac{\mu_0}{\sigma_0^2} + \frac{b_i(y_{i1}-\mu_i)}{\sigma^2}\right).$$

This update balances the prior information about the initial state (through μ_0, σ_0^2) with the information carried by the first observation y_{i1} (through the slope b_i and noise variance σ^2). When $|b_i|$ is large or σ^2 is small, y_{i1} strongly informs y_{i0}; when σ_0^2 is small, the prior dominates and the chain anchors near μ_0.

Appedix J: CPDs Used in the Gibbs Sampler

Table 1: Full Conditional Distributions Used in the Gibbs Sampler

Parameter	**Conditional Distribution**	**Remarks**
$\underset{\sim}{\theta}$	$\mathcal{N}(\mu_\theta, \Sigma_\theta)$	Integrates out $\underset{\sim}{\nu}$
$\underset{\sim}{\nu}$	$\mathcal{N}(V_0^{-1}V_a, V_0^{-1})$	Conditioned on sampled $\underset{\sim}{\theta}$
σ^2	Inverse-Gamma $\left(\frac{T\ell}{2}, \frac{K}{2}\right)$	Sufficient statistic $K = \|y - X\theta - Z\underset{\sim}{\nu}\|^2$
τ_1^2	Inverse-Gamma $\left(\frac{\ell}{2}, \frac{\underset{\sim}{\nu}'(D-\rho_1 W)\underset{\sim}{\nu}_1}{2}\right)$	Prior for spatial variance of $\underset{\sim}{\nu}_1$
τ_2^2	Inverse-Gamma $\left(\frac{\ell}{2}, \frac{(\underset{\sim}{\nu}_2 - A\underset{\sim}{\nu}_1)'(D-\rho_2 W)\underset{\sim}{\nu}_2}{2}\right)$	Prior for spatial variance of $\underset{\sim}{\nu}_2$
ρ_s	Sampled via grid over $\left(\frac{1}{\lambda_{(1)}}, \frac{1}{\lambda_{(\ell)}}\right)$	CAR autocorrelation for $\underset{\sim}{\nu}_s$
$\underset{\sim}{m}$	$\mathcal{N}(G^{-1}\underset{\sim}{h}, G^{-1})$	Prior for latent state
$\underset{\sim}{y}_0$	$\mathcal{N}(A^{-1}B, A^{-1})$ pointwise across ℓ areas	Initial latent observation